Generalized (2+1)d cluster state

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Abstract

In this note we study the generalized cluster state in (2+1)d, which displays a generalized symmetry in both the categorical- and in the higher-form sense. We explicitly present the state in three ways: (1) construction by means of a constant-depth unitary circuit, (2) as the ground state of a commuting stabilizer Hamiltonian, and (3) as a PEPS representation. We distinguish the G-cluster state from the trivial product state by means of a duality/gauging argument.

1 Introduction

2 Notations & conventions

2.1 Qubits $\to$ G-qudits

We generalize the qubits $\{|0\rangle, |1\rangle\}$, the $\mathbb{Z}_2$-case, to G-valued qudits $\{|g\rangle \,|g \in G\}$, such that local Hilbert spaces is the group algebra $\mathbb{C}[G]$, labelled by elements of the finite group G. We denote the inverse of an eleent $g \in G$ as $\bar{g}$. The dual basis is labeled by elements of the irreps, defined as:

$$|\Gamma_{\alpha\beta}\rangle \coloneqq \sqrt{\frac{d_\Gamma}{|G|}} \sum_{g\in G} [\Gamma(g)]_{\alpha\beta} \,|g\rangle \,. \tag{1}$$

Specializing to the trivial irrep yiels:

$$|\mathbb{I}\rangle_{\alpha\beta} = \sqrt{\frac{1}{|G|}} \sum_{g\in G} \delta_{\alpha,\beta} \,|g\rangle \,. \tag{2}$$

2.2 Generalized Pauli Operators

As a reminder we note that the left- and right regular multiplication, which generalizes the Pauli X-operator (i.e. the Pauli σ^x-matrix) is given by:

$$\overrightarrow{X}_g \equiv L_g = \sum_{h\in G} |gh\rangle\langle h| \iff L_g |h\rangle = |gh\rangle . \tag{3}$$

$$\overleftarrow{X}_g \equiv R_g = \sum_{h\in G} |h\rangle\langle hg| \iff R_g |h\rangle = |h\bar{g}\rangle \tag{4}$$

The generalized Pauli Z-operators Z_Γ are given by:

$$Z_\Gamma = \sum_{g\in G} \Gamma(g) \otimes |g\rangle\langle g| \iff Z_\Gamma |g\rangle = \Gamma(g) \otimes |g\rangle \tag{5}$$

$$Z_\Gamma^\dagger = \sum_{g\in G} \Gamma(\bar{g}) \otimes |g\rangle\langle g| \iff Z_\Gamma^\dagger |g\rangle = \Gamma(\bar{g}) |g\rangle \tag{6}$$

The commutation relations of the generalized Pauli-operators acting on the same site read:

$$L_g Z_\Gamma = \Gamma(\bar{g}) \cdot Z_\Gamma L_g \tag{7}$$

$$R_g Z_\Gamma = Z_\Gamma \cdot \Gamma(g) R_g \tag{8}$$

Furthermore, we diagramatically present the generalized Pauli-operators as matrix product operators (MPOs):

(9)

(10)

The diagonal legs indicate physical legs, whilst horizontal- and vertical legs (not shown here) denote virtual legs of the tensors. Note that the virtual legs of the L_g- and R_g- operators are not drawn due to the fact that these MPOs have bond dimension 1.

3 G-Cluster State

We begin by constructing the generalization of the cluster state by replacing the group $\mathbb{Z}_2$ by an arbitrary finite (possibly non-Abelian) group G. We present complementary perspectives on this state: first we present an explicit construction of the state through a constant-depth local unitary quantum circuit (3.1), then we present the state as the unique ground state of the G-cluster commuting stabilizer Hamiltonian (3.2), and finally we present the Projected Entangled-Pair State (PEPS) construction of this state (3.3). In subsection (3.4) we discuss the symmetries of the G-cluster Hamiltonian, and find that it has a $2\mathrm{Vec}_G \boxtimes 2\mathrm{Rep}(G)$-symmetry. Occasionaly, we shall consider the case where $G = \mathbb{Z}_2$.

3.1 Circuit Construction

Our circuit construction closely follows that of Brell [Bre15], inspired by the generalization of the toric code to Kitaev's quantum double models. The first obstacle when considering the generalization is the observation that the $\mathbb{Z}_2$-cluster state is not a CSS stabilizer code (the stabilizers have mixed X- and Z-operators). To this end, one requires applying a Hadamard gate to one of the sublattices of the bipartite graph Λ on which the model is defined. Furthermore, it is not immediately clear if one should replace the Pauli X-operators by either left- or right regular representations of the group G. This problem is mended by requiring that the lattice upon which the model is defined, also be directed $\rightarrow$.

(11)

Explicitely, consider a directed bipartite planar graph $(\Lambda, \rightarrow)$, depicted in eq.(11), and an arbitrary finite group G. We further distinguish between edges $\mathsf{e} \in E \subset \Lambda$ and vertices $\mathsf{v} \in V \subset \Lambda$. Put a qudit v or e on each vertex $\mathsf{v} \in \mathsf{V}$ and each edge $\mathsf{e} \in \mathsf{E}$, respectively, with local Hilbert space isomprphic to the group algebra of G: $\mathcal{H}_{v/e} \cong \mathbb{C}[G]$. The data $(\Lambda, \rightarrow)$ admits a full specification of the CSS G-cluster state $|\mathcal{C}\rangle$ through the circuit construction:

$$|\mathcal{C}\rangle \sim \prod_{\mathsf{e}\in E} \overrightarrow{\prod_{\mathsf{v}\sim\mathsf{e}}} CX_{(v,e)} \, |\mathbb{I}\rangle^{\otimes V} \, |e\rangle^{\otimes E} =: \mathbb{U}_{\mathcal{C}} \, |\mathbb{I}\rangle^{\otimes V} \, |e\rangle^{\otimes E} \tag{12}$$

That is, start with the product state $|\mathbb{I}\rangle^{\otimes V} |e\rangle^{\otimes E}$, where $\mathbb{I} = \frac{1}{|G|} \sum_{g\in G} |g\rangle$, $e \in G$ is the trivial group element, and the tensor product is over all vertices $\mathsf{v} \in V$ and edges $\mathsf{e} \in E$. Subsequently, apply controlled multiplication operators $CX_{(v,e)}$ with as control the vertex qudit v and as target the edge qudit e neighbouring v. Here, we make the choice of applying $C\overrightarrow{X}_{(v,e)} \equiv CL_{(v,e)}$ when $\mathsf{v} = s(\mathsf{e})$. Conversely, we apply $C\overleftarrow{X}_{(v,e)} \equiv CR_{(v,e)}$ when the vertex $\mathsf{v} = t(\mathsf{e})$[1].
In this note, we shall consider a square lattice $\Lambda_\square$, with all horizontal edges directed from left to right, and all vertical edges directed from the bottom of the page towards the top of the page. We use the same convention for multiplication specified in the previous paragraph. The G-cluster state in our convention is given by:

$$|\mathcal{C}\rangle \sim \prod_{\mathsf{e}} CL_{(s(\mathsf{e}),e)} CR_{(t(\mathsf{e}),e)} \, |\mathbb{I}\rangle^{\otimes V} \, |e\rangle^{\otimes E} \, , \tag{13}$$

Thus we can construct the G-cluster state $|\mathcal{C}\rangle$ by means of a constant-depth local unitary quantum circuit from the product state $|\mathbb{I}\rangle^{\otimes V} |e\rangle^{\otimes E}$.

[1] Note that we differentiate between the vertices v and edges e of the lattice, and the vertex-v and edge qudits e.

3.2 Ground State of a Stabilizer Hamiltonian

We generalize the (2+1)d $\mathbb{Z}_2$-cluster state Hamiltonian [CSSZ25][VBV+24][Yos16] by replacing Pauli-operators with generalized Pauli-operators following Fechisin [FTA25]. We again require a directed bipartite lattice as in section (3.1).
The (2+1)d (CSS) G-cluster state Hamiltonian then reads:

$$\mathbb{H}_{\mathcal{C}}^{G} = -\frac{1}{|G|}\sum_{\mathsf{v}\in\mathsf{V}}\sum_{g\in G} A_g^{\mathsf{v}} - \frac{1}{|G|}\sum_{\mathsf{e}\in\mathsf{E}}\sum_{\Gamma\in\mathrm{Rep}(G)} B_{\Gamma}^{\mathsf{e}} \tag{14}$$

where the vertex-centered- A_g^{v} and edge-centered operators B_{Γ}^{e} are given by:

$$A_G^{\mathsf{v}} \coloneqq \mathrm{Star}_g^{\mathsf{v}} \tag{15}$$

$$B_{\Gamma}^{\mathsf{e}} \coloneqq d_{\Gamma}\mathrm{Tr}[Z_{\Gamma}^{s(\mathsf{e})}\cdot Z_{\Gamma}^{(\mathsf{e})}\cdot Z_{\Gamma}^{t(\mathsf{e})\dagger}] \tag{16}$$

The first summation in eq.14 puts *star operators* on each vertex, the form of which depends on the choice of directionality of the lattice. The second summation puts 3-site operators on all edges, similar to the $(1+1)$d G-cluster Hamiltonian [FTA25]. Notice that the Hamiltonian $\mathbb{H}_{\mathcal{C}}^{G}$ does not commute with the G-cluster state preparation unitary circuit $\mathcal{U}_{\mathcal{C}}$. We note that for the square lattice $\Lambda_{\square}$ with arrows pointing left-to-right and bottom-to-top, the star operators centered on vertex $\mathsf{v}\in V$ connected to edges $\mathsf{e}\in E$ are of the form:

$$\tag{17}$$

The individual terms are interpreted as stabilizers of the G-cluster state. One can show that the G-cluster state is the unique ground state (GS) of the Hamiltonian $\mathbb{H}_{\mathcal{C}}^{G}$ by counting the degrees of freedom on the lattice and the number of stabilizers contained in $\mathbb{H}_{\mathcal{C}}^{G}$. We also provide a Hamiltonian $\mathbb{H}_{\mathrm{TRIV}}^{G}$, to which one can compare $\mathbb{H}_{\mathcal{C}}^{G}$ to:

$$\mathbb{H}_{\mathrm{TRIV}}^{G} \coloneqq -\frac{1}{|G|}\sum_{v}\sum_{g\in G} L_g^{(v)} - \frac{1}{|G|}\sum_{\mathsf{e}}\sum_{\Gamma\in\mathrm{Rep}(G)} \mathrm{Tr}[Z_{\Gamma}^{(\mathsf{e})}]d_{\Gamma}, \tag{18}$$

which has as GS the product state $|\psi_0^{\mathrm{TRIV}}\rangle = |\mathbb{I}\rangle^{\otimes E}|e\rangle^{\otimes V}$.
For the case where $G=\mathbb{Z}_2$, the generalized Pauli Z- and X-operators (Z_{Γ}, and R_g or L_g) reduce to the usual Pauli matrices Z and X. The Hamiltonian $\mathbb{H}_{\mathcal{C}}^{G}$ becomes:

$$\mathbb{H}_{\mathcal{C}}^{\mathbb{Z}_2} = -\sum_{\mathsf{v}}\mathsf{Star} - \sum_{\mathsf{e}} Z^{s(\mathsf{e})}Z^{(\mathsf{e})}Z^{t(\mathsf{e})}, \tag{19}$$

wherein we ommited the trivial stabilizers. Here Star is of the same form as eq. 17, but with Pauli X-operators only.

3.3 G-Cluster State PEPS

(20)

(21)

We now present the PEPS representation [CPGSV21] of the G-cluster state, which we depict in eq.(**??**). The horizontal- and vertical legs of each tensor indicate virtual legs (coloured blue), whereas diagonal legs of each tensor indicate physical legs (coloured black). The individual tensors[2] are given by:

(22)

In order to establish that eq.(**??**) is the correct PEPS representation, it suffices to check that it stabilizes, i.e. it is in the $+1$ eigenspace, of all stabilizers contained in $\mathbb{H}_{\mathcal{C}}^{G}$. To this end, we give the symmetries of the vertex- and edge tensors:

(23)

(24)

(25)

(26)

(27)

(28)

It should be clear from the above symmetries that the PEPS representation (eq.**??**) is indeed the GS of $\mathbb{H}_{\mathcal{C}}^{G}$. Indeed, pulling the stabilizers through to the virtual legs, one get the identity operators with $+1$ eigenvalue. Take for example

[2] We proceed to write first the virtual legs of the tensor, followed by the physical leg of the tensor.

a ZZZ-operator: on the vertices one pulls the first $Z_\Gamma^\dagger$ throught toward the left virtual leg of the edge tensor and the last Z_Γ throught towards the right virtual leg of the edge tensors, meanwhile one pulls the middle Z_Γ through which yields a Z_Γ and a $Z_\Gamma^\dagger$ on the left and right virtual legs off the edge tensor, respectively. What remains is the identity, with trace equal to d_Γ, such that the full term yields eigenvalue $+1$. A similar argument can be made for the terms in the second sum.

3.4 Symmetries of the G-cluster Hamiltonian

In this subsection, we analyse the symmetries of the G-cluster state. Specifically, we find that the state has the following independent families of global symmetries:

$$\mathcal{V}_g^{(0)} := \prod_{\mathsf{v}\in V} R_g^{\mathsf{v}} \tag{29}$$

$$\mathcal{U}_\Gamma^{(1)}(\gamma) := \mathrm{Tr}[\overrightarrow{\prod_{\mathsf{e}\subset\gamma}} Z_\Gamma^{\mathsf{e}}], \tag{30}$$

where γ is some closed oriented loop defined on the dual lattice $\Lambda^\star$, and which places operators on the edges e intersecting with γ. The $\overrightarrow{\prod}$-product indicates a oriented product, wherein we use the following convention:

$\equiv Z_\Gamma$ $\equiv Z_\Gamma^\dagger$ (31)

Importantly, we note that the symmetries $\mathcal{V}$ and $\mathcal{U}$ act on different sublattices, such that they commute with one another. The superscripts (0) and (1) indicate that we are dealing with a (conventional) 0-form, and a 1-form symmetry, respectively. Furthermore, since multiplication of irreps do not generally multiply according to the rules of a group, we have that the $\mathcal{U}$-symmetry is also a non-invertible symmetry, multiplying according to the rules of the fusion category $\mathrm{Rep}(G)$. We thus have that the $\mathcal{U}$-symmetry generalizes the concept of symmetries in two ways: both in the categorical sense, and in the higher-form sense [FTA25]. Furthermore, $\mathcal{U}$ is topological, since we can safely deform the loop $\gamma \mapsto \gamma'$. This fact is a direct condition of initializing the gauge fields to be *flat* upon gauging a SSB model. This is elaborated upon in appendix 4.

4 Duality: $2\mathbf{Vec}_G \boxtimes 2\mathbf{Vec}_G$ SSB $\to 2\mathbf{Vec}_G \boxtimes 2\mathbf{Rep}(G)$ SPT

In this section we give a duality argument, similar to the (1+1)d case in [FTA25]. We will show that the G-cluster state is distinct from the trivial product state

by showing that upon gauging/ dualising models with different spontaneous symmetry breaking (SSB) patterns, we obtain the aforementioned states as unique GSs of the dual models. Consider the following two models defined on the directed lattice Λ' consisiting of two-component vertices labeled $\mathsf{v} \equiv (\mathsf{v}_1, \mathsf{v}_2) \in \mathsf{V}$:

$$\mathbb{H}_{\mathrm{SSB1}} = -\frac{1}{|G|}\sum_{\mathsf{v}}\sum_{g\in G} L_g^{\mathsf{v}_1} - \frac{1}{|G|}\sum_{\mathsf{e}}\sum_{\Gamma\in\mathrm{Rep}(G)} d_\Gamma \mathrm{Tr}[Z_\Gamma^{s(\mathsf{e})_2}\cdot Z_\Gamma^{t(\mathsf{e})_2\dagger}], \tag{32}$$

$$\mathbb{H}_{\mathrm{SSB2}} = -\frac{1}{|G|}\sum_{\mathsf{v}}\sum_{g\in G} L_g^{\mathsf{v}} - \frac{1}{|G|}\sum_{\mathsf{e}}\sum_{\Gamma\in\mathrm{Rep}(G)} d_\Gamma \mathrm{Tr}[Z_\Gamma^{s(\mathsf{e})_1}\cdot Z_\Gamma^{s(\mathsf{e})_2}\cdot Z_\Gamma^{t(\mathsf{e})_2\dagger}\cdot Z_\Gamma^{t(\mathsf{e})_1\dagger}], \tag{33}$$

Note that $L_g^{\mathsf{v}} \equiv L_g^{\mathsf{v}_1} L_g^{\mathsf{v}_2}$. Both of the model Hamiltonians $\mathbb{H}_{\mathrm{SSB1}}$ and $\mathbb{H}_{\mathrm{SSB2}}$ have the following families of symmetries which act on the first- and second components of each vertex $\mathsf{v} \in \Lambda'$, respectively:

$$G_I : \prod_{\mathsf{v}} R_g^{\mathsf{v}_1} \tag{34}$$

$$G_{II} : \prod_{\mathsf{v}} R_g^{\mathsf{v}_2} \tag{35}$$

Since these symmetries act indepentently, the total symmetry is $G_I \times G_{II}$. In both models, the GS spontaneously breaks symmetry. For SSB1 the GS is given[3] by $|\psi_0^{\mathrm{SSB1}}\rangle \coloneqq |\mathbb{I}\rangle^{\otimes \mathrm{V1}} |g\rangle^{\otimes V2}$ for $g \in G$, with unbroken symmetry G_I. For SSB2 the GS is given by $|\psi_0^{\mathrm{SSB2}}\rangle \coloneqq \prod_{\mathsf{e}} CL_{(s(\mathsf{e})_1, t(\mathsf{e})_2)} CR_{(t(\mathsf{e})_1, s(\mathsf{e})_2)} |\mathbb{I}\rangle^{\otimes V1} |g\rangle^{\otimes V2}$ for $g \in G$, which has unbroken symmetry[4] $\{\prod_{\mathsf{e}} R_h^{s(\mathsf{e})_1} R_{\bar{g}hg}^{t(\mathsf{e})_2} L_h^{t(\mathsf{e})_1} L_{g\bar{h}\bar{g}}^{s(\mathsf{e})_2} | h \in G\}$ for the state labeled by g. Comparing the GSs, we see that they are related by the circuit:

$$U_{\mathrm{SSB}} \coloneqq \prod_{\mathsf{e}} CL_{(s(\mathsf{e})_1, t(\mathsf{e})_2)} CR_{(t(\mathsf{e})_1, s(\mathsf{e})_2)} \tag{36}$$

such that $|\psi_0^{\mathrm{SSB2}}\rangle = U_{\mathrm{SSB}} |\psi_0^{\mathrm{SSB1}}\rangle$ and $U_{\mathrm{SSB}} \mathbb{H}_{\mathrm{SSB1}} U_{\mathrm{SSB}}^\dagger = \mathbb{H}_{\mathrm{SSB2}}$.

We now apply the gauging procedure outlined in [HVAS+15] to the above models. Specifically, we gauge the G_{II}-symmetry. This is done by first introducing edge degrees of freedom (d.o.f.) i.e. G-qudits on the edges between the sites $\mathsf{v} \in \Lambda'$. We refer to the original sites as the matter d.o.f., and the newly introduced edge d.o.f. as gauge d.o.f.. Then we apply a group-averaged projector $\mathcal{P} = \frac{1}{|G|}\sum_{g\in G} \mathcal{P}_g$, projecting onto the locally gauge-invariant subspace. $\mathcal{P}_g$ is interpreted as a (local) Gauss law. We supplement the gauging projector by

[3] We denote by $V1$ all the first components v_1 for each vertex $\mathsf{v} \in V \subset \Lambda'$. Similarly, we denote by $V2$ all the second components.

[4] Ambiguity of the ordering is resolved by fixing the ordering here to be clockwise around any given vertex $\mathsf{v} \in V$.

a *disentangler* $\mathcal{D}$. The disentangler, due to totally gauging the G_{II}-symmetry, decouples the (second component of the) matter d.o.f. from the gauge d.o.f.. We defer details of the procedure to appendix A. For our purpose, we only mention the following results:

$$Z_\Gamma^{s(\mathsf{e})} \cdot Z_\Gamma^{t(\mathsf{e})\dagger} \mapsto Z_\Gamma^{\mathsf{e}}, \tag{37}$$

$$L_g^{\mathsf{v}} \mapsto \prod_{s(\mathsf{e})=\mathsf{v}} L_g^{\mathsf{e}} \prod_{t(\tilde{\mathsf{e}})=\mathsf{v}} R_g^{\tilde{\mathsf{e}}}. \tag{38}$$

Applying the above results to the SSB Hamiltonians (upon a relabelling of $\mathsf{v}_1 \mapsto \mathsf{v}$) yields:

$$\mathbb{H}_{\text{SSB1}} \mapsto -\frac{1}{|G|}\sum_v \sum_{g\in G} L_g^{(v)} - \frac{1}{|G|}\sum_{\mathsf{e}} \sum_{\Gamma\in\text{Rep}(G)} d_\Gamma \text{Tr}[Z_\Gamma^{(\mathsf{e})}] \tag{39}$$

$$= \mathbb{H}_{\text{TRIV}}^G \tag{40}$$

$$\mathbb{H}_{\text{SSB2}} \mapsto -\frac{1}{|G|}\sum_{\mathsf{v}} \sum_{g\in G} \mathsf{Star}_g^{\mathsf{v}} - \frac{1}{|G|}\sum_{\mathsf{e}} \sum_{\Gamma\in\text{Rep}(G)} d_\Gamma \text{Tr}[Z_\Gamma^{s(\mathsf{e})} \cdot Z_\Gamma^{(\mathsf{e})} \cdot Z_\Gamma^{t(\mathsf{e})\dagger}] \tag{41}$$

$$= \mathbb{H}_{\mathcal{C}}^G \tag{42}$$

$$\tag{43}$$

Hence, we find that SSB1 is mapped to a model with unique (product) GS $|\Psi_0^{SSB1}\rangle := |\mathbb{I}\rangle^{\otimes E} |e\rangle^{\otimes V}$. Meanwhile, the SSB2 Hamiltonian $\mathbb{H}_{\text{SSB2}}$ is mapped to the G-cluster state Hamiltonian $\mathbb{H}_{\mathcal{C}}^G$, with unique ground state the (2+1)d G-cluster state $|\mathcal{C}\rangle$, which is a SPTO state. Finally, we note the similarity between the circuit $\mathbb{U}_{\mathcal{C}}$ used to prepare the G-cluster state in subsection 3.1 and U_{SSB}, which have equivalent (in the sense of the gauging map above) controls, and targets (even sites in the SSB model, which are mapped to edge sites in the SPT model).

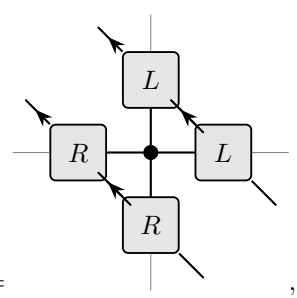
Appendix A: Gauging 2Vec_G

We follow the gauging procedure at the level of states outlined in [HVAS+15]. We illustrate the gauging of G_{II} defined in section 4. Since G_{II} only acts upon the second components of each vertex $\mathsf{v} \equiv (\mathsf{v}_1, \mathsf{v}_2)$, we temporarily 'forget' the first components $\mathsf{v}_1 \quad \forall \mathsf{v} \in V$, and add these back in at the end of the dualitisation procedure. We use the term *dualisation* as the result of *gauging* appended by *disentangling*. The disentangler, to be introduced below, essentially seeks to decouple the gauge d.o.f. from the matter d.o.f..
Given the directed lattice Λ', denote by E and V the sets consisting of all the

edges e and vertices v, respectively. For a specific edge e, the arrow on this edge defines the source $s(\mathsf{e})$ and target $t(\mathsf{e})$. Given the finite (possibly non-Abelian) group G , introduce on each edge $\mathsf{e} \in \mathsf{E}$ a qudit e with local Hilbert space the group algebra $\mathbb{C}[G]$ such that an arbitrary single-site state may be written as $|\phi\rangle = \sum_{g\in G} c_g |g\rangle$ where $c_g \in \mathbb{C}$. We refer to the newly introduced qudits on the edges as *gauge* d.o.f..
Next, we set up the group-averaged projector $\mathcal{P}^{\mathsf{v}} = \frac{1}{|G|}\sum_{g\in G} \mathcal{P}^{\mathsf{v}}_g$, $\mathcal{G} := \bigotimes_{\mathsf{v}\in\mathsf{V}} \mathsf{P}^{\mathsf{v}}$. For the case where Λ' is a square lattice with all edges directed left-to-right and bottom-to-top, $\mathcal{P}^{\mathsf{v}}_g$ reads:

(44)

with $u_g = R_g$ [HVAS+15]. The $\mathcal{P}^{\mathsf{v}}_g$ locally enforce a *Gauss law*. We are free to choose any $g \in G$ to initialize the gauge d.o.f., and choose the trivial group element $e \in G$ for simplicity. The distinction between edges and the trivial group element should always be clear from the context. If a $g \neq e$ is chosen, one simply appends an extra layer to the quantum circuit implementing the duality to correct this lapse in judgement.
We now introduce the disentangler $\mathcal{D} := \bigotimes_{\mathsf{v}\in V} \mathcal{D}^{\mathsf{v}}$. There is no clear-cut formula to determine the form of the disentangler. Inspired by the duality procedure used for the duality argument for the (1+1)d G-cluster state in [FTA25], we find the following disentangler for the square lattice Λ':

$\mathcal{D}^{\mathsf{v}} :=$ 

, (45)

centered on v and consisting of generalized CX gates. The gates have as common control the vertex v and as targets the qudits e on the adjacent edges e. Importantly, we make the following observation:

$$\mathcal{D}^{\mathsf{v}}\mathcal{P}^{\mathsf{v}}_g = R^{\mathsf{v}}_g\mathcal{D}^{\mathsf{v}} \implies \mathcal{D}^{\mathsf{v}}\mathcal{P}^{\mathsf{v}} = \frac{1}{|G|}\sum_{g\in G} R^{\mathsf{v}}_g\mathcal{D}^{\mathsf{v}} := \mathcal{T}^{\mathsf{v}}\mathcal{D}^{\mathsf{v}}, \tag{46}$$

wherein we have defined the *trivializer* $\mathcal{T}^{\mathsf{v}}$ which maps the vertex state to $|\mathbb{I}\rangle$. In summary, we first applied the gauging projector $\mathcal{P}$ and subsequently applied the disentangler $\mathcal{D}$. The observation in eq. 46 significantly simplifies the quantum circuit implementing the duality, by trivializing the vertex state.

Eqs. 37 can now be rewritten in terms of the *duality* operator $\mathbb{D}$ [SS24]:

$$\mathbb{D}\Big[Z_\Gamma^{s(\mathsf{e})}\, Z_\Gamma^{t(\mathsf{e})\dagger}\Big] = Z_\Gamma^{\mathsf{e}}\, \mathbb{D} \tag{47}$$

$$\mathbb{D}\big[L_g^{\mathsf{v}}\big] = \left[\prod_{s(\mathsf{e})=\mathsf{v}} L_g^{\mathsf{e}} \prod_{t(\bar{\mathsf{e}})=\mathsf{v}} R_g^{\bar{\mathsf{e}}}\right] \mathbb{D}. \tag{48}$$

The gauge d.o.f.s satisfy the so called *flatness condition* around each plaquete. By *flatness* we mean that an appropriate product of the gauge d.o.f. yield the group identity element. For example, for the choice of orientation of the square lattice given in section 3.1, taking the product of the top edge-, the inverse of the right edge-, the inverse of the bottom edge-, and the left edge of a plaquette, yields $e \in G$. This is a direct consequence of the fact that the gauge d.o.f. at the start of the gauging procedure above were all initialized to the same state, more specifically the trivial state, together with the fact that each edge connects two neighbouring vertices. This can also be seen by looking at the gauging map (which first introduced the gauge d.o.f., and subsequently applies the group-averaging projector on all sites), whicht acts on an arbitrary initial state $|\psi\rangle$:

$$\mathcal{G}\,|\psi\rangle = \prod_{\mathsf{v}\in\mathsf{V}} \frac{1}{|G|} \sum_{g\in G} u_g^{\mathsf{v}}\,|\psi\rangle \bigotimes_{\mathsf{e}\in\mathsf{E}} |g_{\mathsf{s}(\mathsf{e})} g_{\mathsf{t}(\mathsf{e})}^{-1}\rangle\,. \tag{49}$$

For any oriented closed loop $l \in Z^1(\Lambda^*, \mathbb{C}[G])$, defined on the dual lattice Λ^*, notice that due to the gauging map projecting the gauge d.o.f. to the flat subspace, we always get $+1$ for the symmetry operator $\mathcal{U}_\Gamma(l)$ (if normalised correctly). The ability to deform the loops $l \mapsto l' \in Z^1(\Lambda^*, \mathbb{C}[G])$ implies that $\mathcal{U}_\Gamma$-operators contained in $2\mathrm{Rep}(G)$ are topological. Furthermore, the following identity holds:

$$\mathcal{U}_\Gamma^{(1)}(\gamma)\mathbb{D} = \mathbb{D} \tag{50}$$

$$\tag{51}$$

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